

Name
Date
Course

Exponential Equations and Logarithmic Equations (Homework)

Solve. Write the Roman numeral of each exponent rule that you use.

$$\textcircled{1} \frac{1}{9^{-x}} = \sqrt[7]{81^3}$$

$$\textcircled{2} (\sqrt{5})^x = \frac{1}{125^{-1}}$$

$$\textcircled{3} \sqrt[4]{100^x} = \frac{1}{10^{-7}}$$

$$\textcircled{4} \frac{1}{8^{5x+4}} = (\sqrt[3]{4})^x$$

Find the value of each logarithm without a calculator.

⑤ $\log_7 49$

⑥ $\log_5 1$

⑦ $\log_3 27$

⑧ $\log_2 16$

Determine if the following logarithms can be found without a calculator. If so, calculate the values. If you use an exponent rule, write its Roman numeral.

⑨ $\log_3 32$

⑩ $\log_4 16$

⑪ $\log_5 2$

⑫ $\log_2 4$

$$\textcircled{13} \log_{25} 5$$

$$\textcircled{14} \log_8 16$$

$$\textcircled{15} \log_3 82$$

$$\textcircled{16} \log_9 25$$

$$\textcircled{17} \log_2 \left(\frac{1}{4}\right)$$

$$\textcircled{18} \log_7 \sqrt[3]{49}$$

Find the value of each logarithm without a calculator.

$$\textcircled{19} \log_6 6^2$$

$$\textcircled{20} \log_9 9$$

$$\textcircled{21} \log_{12} 12^{10}$$

$$\textcircled{22} \log_5 5^7 \quad \textcircled{23} 8^{\log_8 3} \quad \textcircled{24} \log_4 4$$

$$\textcircled{25} 3^{\log_3 4} \quad \textcircled{26} \log_7 7 \quad \textcircled{27} 5^{\log_5 \left(\frac{2}{3}\right)}$$

Solve without a calculator. If you use an exponent rule, write its Roman numeral.

$$\textcircled{28} \log_x 16 = 2 \quad \textcircled{29} x = \log_{81} 27 \quad \textcircled{30} \log_4 x = 3$$

$$\textcircled{31} -1 = \log_6 x \quad \textcircled{32} \log_x 81 = 4 \quad \textcircled{33} \log_{36} 6 = x$$

$$\textcircled{34} x = \log_2 \sqrt{8} \quad \textcircled{35} -\frac{1}{3} = \log_8 x \quad \textcircled{36} 5 = \log_x 32$$

Exponential Equations and Logarithmic Equations (Homework) Answers

① $x = \frac{6}{7}$

② $x = 6$

③ $x = 14$

④ $x = -\frac{36}{47}$

⑤ 2

⑥ 0

⑦ 3

⑧ 4

⑨ Calculator required

⑩ 2 , No calculator required

⑪ Calculator required

⑫ 2 , No calculator required

⑬ $\frac{1}{2}$, No calculator required

⑭ $\frac{4}{3}$, No calculator required

⑮ Calculator required

⑯ Calculator required

⑰ -2 , No calculator required

⑱ $\frac{2}{3}$, No calculator required

⑲ 2

⑳ 1

㉑ 10

㉒ 7

㉓ 3

㉔ 1

㉕ 4

㉖ 1

㉗ $\frac{2}{3}$

㉘ $x = 4$, $x \neq -4$

㉙ $x = \frac{3}{4}$

㉚ $x = 64$

㉛ $x = \frac{1}{6}$

㉜ $x = 3$, $x \neq -3$

㉝ $x = \frac{1}{2}$

㉞ $x = \frac{3}{2}$

㉟ $x = \frac{1}{2}$

㊱ $x = 2$